



VinFast VF9 EV SUV

First Responder Guide

Legal

Legal notice:

This guide was created only for first and second responders who have specific training in the area of technical assistance following road accidents, and are therefore able to carry out the activities described in this guide. Specifications and optional equipment for VinFast vehicles and the range of vehicles made by VinFast are subject to ongoing changes. VinFast therefore explicitly reserves the right to make changes or modifications to this guide at any time. The information factors in the state of knowledge at the date of creation.

Please note:

The information in this guide is not intended for end customers, nor is it intended for workshops and dealerships. End customers can find information about the functions of their vehicles and important safety information on vehicle and occupant safety in the owner's manual of the respective VinFast vehicle. Workshops and authorized dealers can obtain repair information from the usual sources known to them.

Contents

This First Responder Guide covers emergency vehicle shutdown information for the VinFast VF9 EV SUV. The VF9 is an all-electric vehicle and contains High Voltage (HV) components. The HV battery is the main energy source. High voltage refers to systems with DC voltage above 60 V or AC voltage above 30 V.

The information in this guide is intended for use only by qualified and trained first responders. This guide describes how to identify HV components and safely handle them during an emergency. It also explains how to disable the HV system and provides safety information specific to the VF9, including high-strength materials used in the vehicle structure. Failure to follow the practices recommended in this guide can result in serious injury or death.

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Warnings

		
DANGER	WARNING	CAUTION
Will result in serious personal injury or death.	May cause serious injury or death.	May cause personal injury or component damage.



Always assume that HV parts are energized. Always wear the appropriate Personal Protective Equipment (PPE). Contact with HV current can cause serious injury or death.

Recovery of a submerged vehicle without appropriate PPE can result in serious injury or death.

Batteries can release flammable or toxic gases if they are damaged. Gases can be ignited by a spark or heat source. Pressure from expanding gases can build up in battery packs and cause an explosion. These situations can lead to serious injury or death.



Do not use the HV battery to lift the vehicle. The HV battery could become damaged. This increases the risk of injury.

If the electric vehicle is on fire, it can be extinguished with large quantities of water. Use water from a hydrant or other nearby water sources whenever possible.

Lithium-ion batteries can spontaneously self-ignite. Batteries can reignite after a fire has been suppressed. This can occur several hours after an event that causes damage or as the result of incorrect use. Always wear the appropriate PPE.

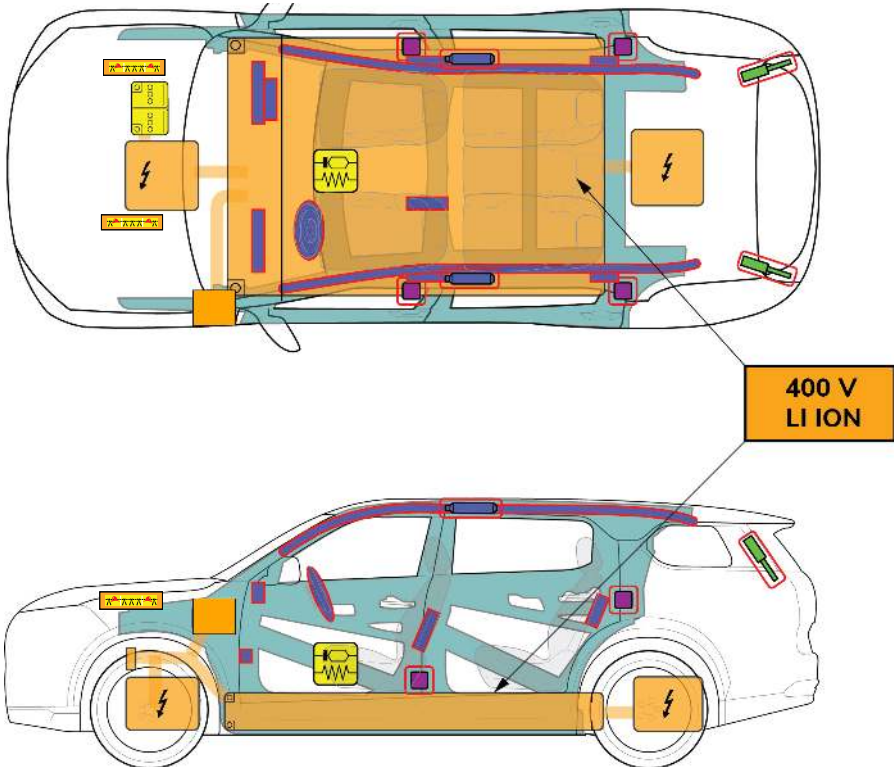
PPE must be worn for all practices discussed in this guide. Appropriate PPE is required for the procedure to disable the HV system. PPE may include rubber gloves, safety glasses, electrical hazard footwear, earplugs, hardhats and face shields, respirators, or arc-rated clothing and undergarments.

0. Rescue Sheet


VINFAST

VF9
 5Doors SUV
 From 2024





Note: The illustration shows the maximum possible range of equipment.

	Airbag		Airbag Inflator/ Stored gas Inflator		Seat belt pretensioners		SRS Control Unit		Gas strut/ Pre loaded Spring
	High Strength Zone		Battery, Low Voltage		Battery Pack, High Voltage		High Voltage		High Voltage Power cable
	Cable Cut								

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1. Identification / recognition

Exterior badges identify the VinFast VF9 EV SUV. Each model has distinguishing features. Individual models can be identified by body shape, size, and vehicle design.



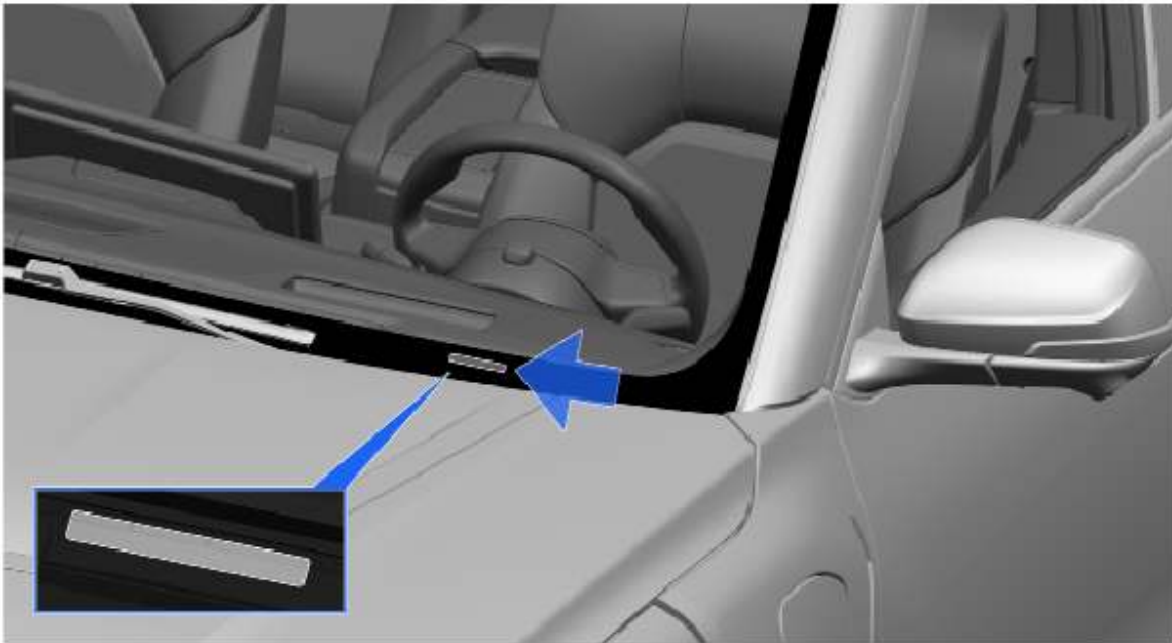
Lack of engine noise does not mean vehicle is off: silent movement or instant restart capability exists until vehicle is fully shut down. Wear appropriate PPE.



Vehicle Identification Number

The VIN is located in three places on the vehicle.

1. Look through the windshield on the driver's side. A stamped plate on the dashboard provides the VIN.
2. Lift the carpet flap to reveal a stamped plate that provides the VIN.
3. On the main display unit, navigate to Settings > About > Vehicle.



Engine Compartment

Engine compartment is accessed by opening the hood. Once hood is opened and secured we will find the front trunk compartment. The front trunk need to be removed to find the different components of engine compartment. See below image.



2. Immobilisation/ stabilisation/ lifting

The Electric engine is noiseless. The display in the touchscreen shows a message as to whether the electric drive is switched "OFF" or ready for operation "READY"

Stabilisation



Wheel rotation generates high voltage and must be prevented while stabilizing the vehicle.

Step 1:

Press the Park (P) button on the driver console to engage the parking brake.



Step 2:

Chock the wheels to prevent movement in case the parking brake fails.



Pushing

The VF9 can be pushed short distances to clear the roadway when safety allows. Press the brake pedal and the "N" button on the driver console to put the vehicle in neutral.

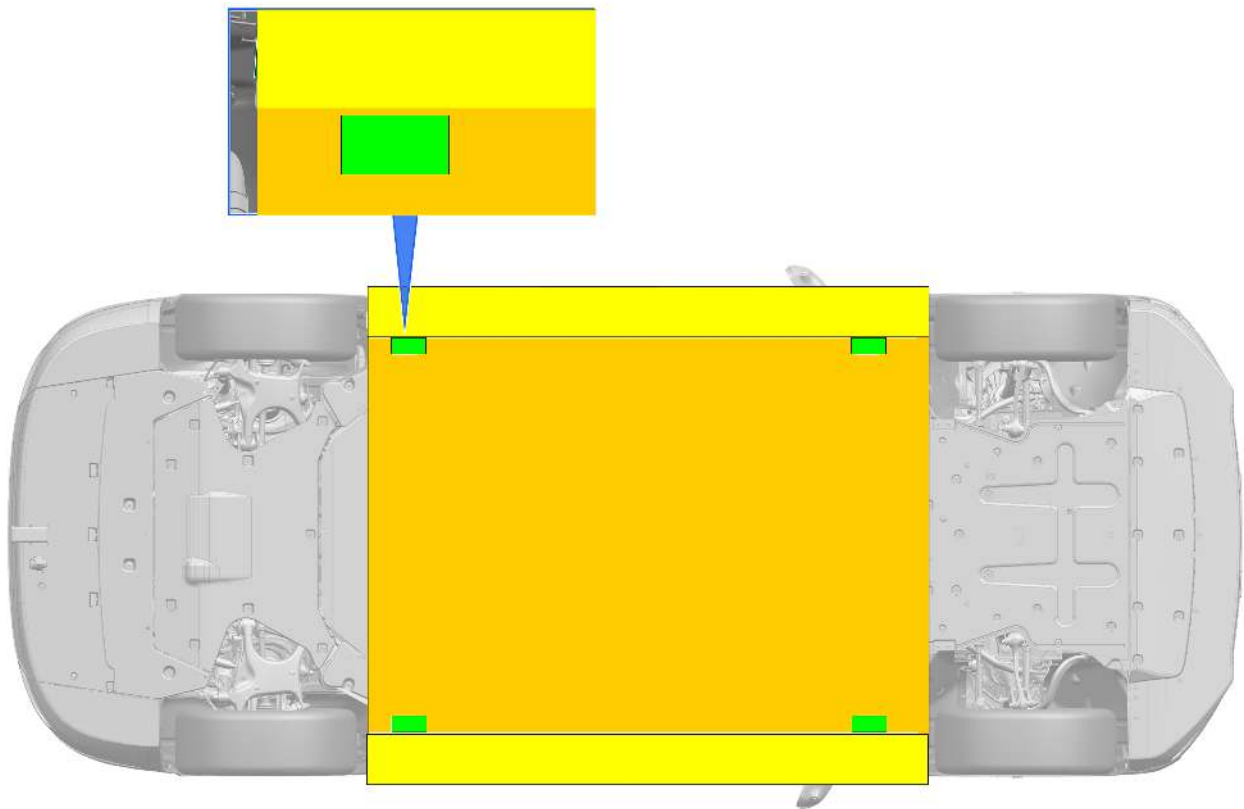


Lifting



Do not use the HV battery to lift the vehicle. The HV battery could become damaged. This increases the risk of injury.

Extreme caution is required when lifting the vehicle. The VF9 has a 400-volt lithium-ion HV battery mounted underneath the floor pan. When lifting or jacking the vehicle, only use the designated lifting points marked in green.



This guidance applies only to non-deformed vehicles. If the VF9 involved in a crash is deformed, first responders must decide how to safely lift the vehicle.

3. Disable direct hazards/ safety regulations

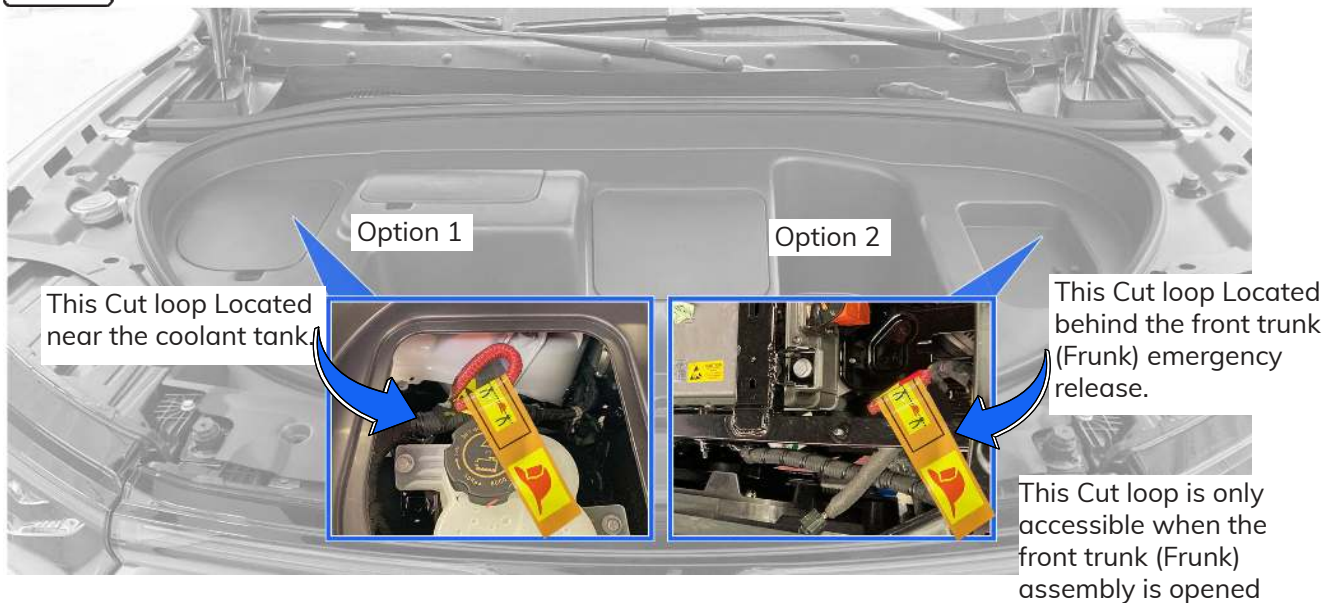
Main Disabling Method

Open the hood. There are two first responder cut loops in the engine compartment. One is located behind the coolant tank (Option 1), and one is located behind the front trunk emergency release (Option 2). The cutloop located behind the frunk requires the removal of the entire front trunk (frunk) assembly and its attaching parts. (See below Image)

Option 1- Lift the plastic flap to access the first responder cut loop near the coolant tank, cut the loop twice.

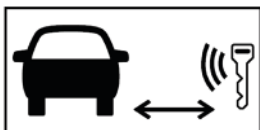
Option 2- Remove the front trunk look for the cut loop label and cut the loop twice.

Cutting one of the loop will interrupt the 12V power to the HV battery, which opens the contactors in the HV battery. Cut loops need to be cut at two places to prevent the wires from accidentally contacting each other.



It takes about 5 minutes for the high voltage to dissipate

Remove Smart Key/ Starter Key

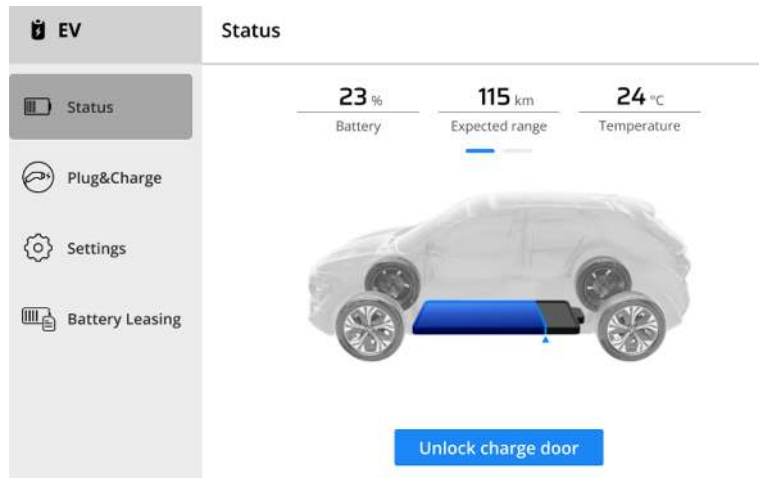


If the key is outside the vehicle, the VF9 will shutdown automatically. Keeping the key at least 19 feet away from the vehicle will prevent accidental operation of electrical systems.

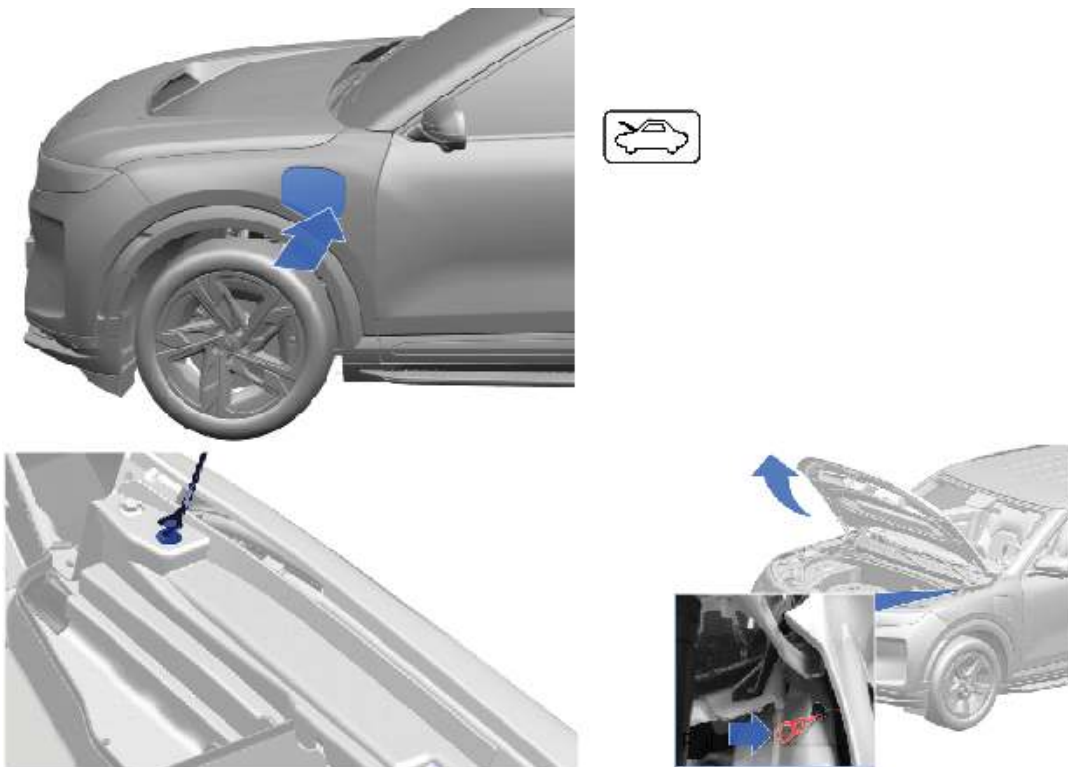
Disabling Method for Service

Step 1:

Open the EV app and go to status, press the Unlock Charge Door button. Then, push the charge port flap once to open it.



To open it manually, open the hood, remove beauty cover and pull a Bowden cable next to the hood hinge to release the flap. Then, push the charge port flap once to open it. For details on opening the hood, see [Chapter 4](#).



A Mechanical Service Disconnect (MSD) is located within the charge port. The MSD is low voltage and interrupts the 12V power supply to the HV battery. Disconnecting the MSD initiates an HV system shutdown.



Step 2:

Remove the MSD as follows



Remove the security flap.



Pull the red safety cap.



Pull the MSD tab.

Disable HV System While Charging

The charging plug must be removed to disable the HV system during charging. The charging plug can only be removed when the vehicle's doors are unlocked. If the charging plug is not removed from the charging port within 60 seconds, charging will resume.

Mechanical emergency release cables are located on the left side of the hood compartment and next to the hood hinge. The charging handle emergency release cable is near the wiper motor.

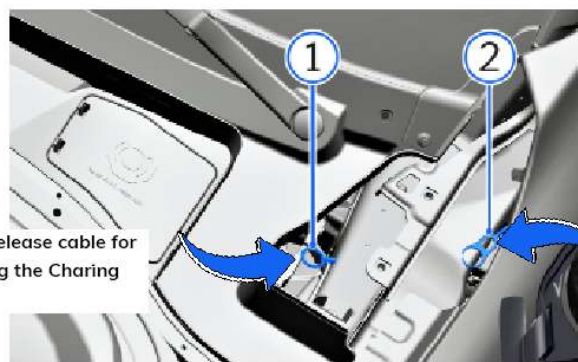
To stop the charging process and release the charging handle locking pin, press "Stop Charging" on the main display unit.

To release the charging plug manually:

1. Press the Stop Charging button on the main display unit inside the vehicle.
2. Pull the mechanical release cable (1) to disconnect the charging plug.



Make sure the vehicle is powered off. Contact with moving wiper motor mechanism may cause serious injury.



Emergency release cable for disconnecting the Charging Plug.



Emergency release cable for opening the Charging Port Door/Flap.

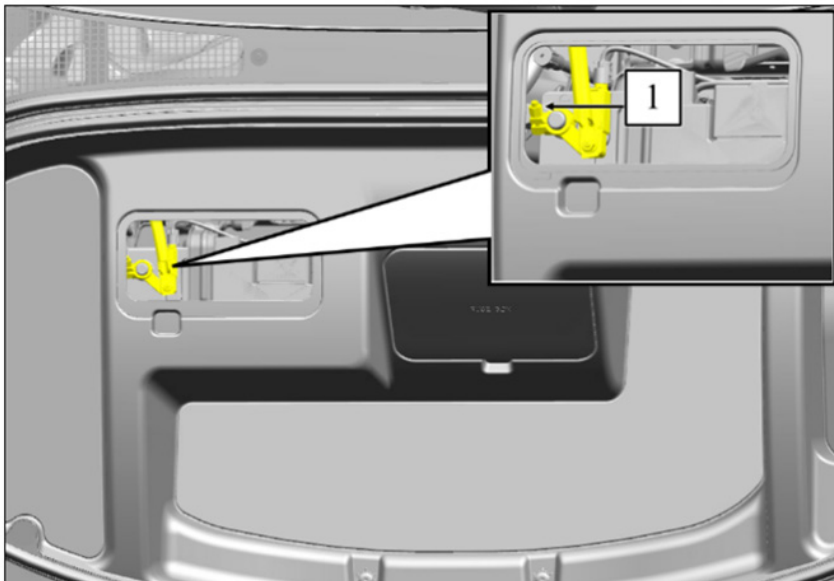
There is a second emergency release cable (2) that opens the charging port flap manually if it is not electrically released.

Disabling LV System

Disconnect the 12V battery by disconnecting the negative terminal of the battery.



Open the hood and then open Maintenance Cap



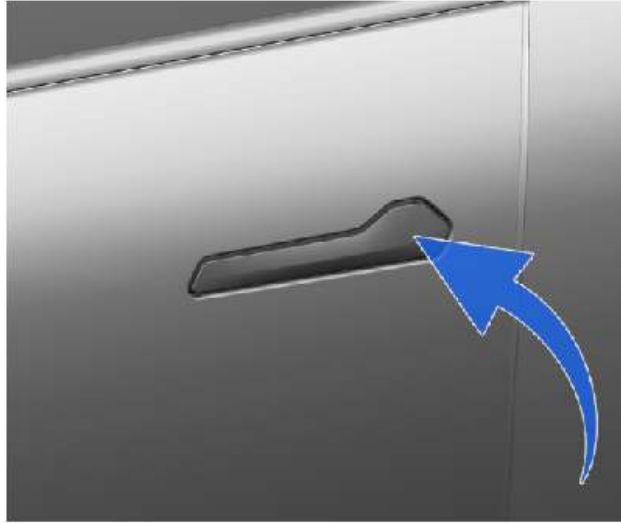
Loosen the screw and pull the battery negative cable.

4. Access to the occupants

Accessing the Vehicle

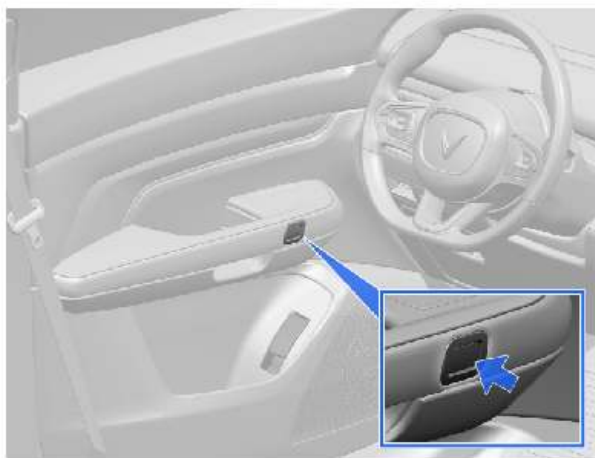
Opening doors from the outside

Push and hold the widest part of the doorhandle. The thinner side of the handle pivots out. Pull the handle to open the door.



Opening doors from the inside with power

Press the button near the door panel to open the door from the inside when 12V power is available



Opening doors from the inside without power

Locate the mechanical release handle near the window operation switches. Lift the mechanical release handle to open the front doors from the inside when 12V power is unavailable.



If the exterior door handles do not work, reach into the window and pull the interior door handle.

Seat Adjustments



Under no circumstances should the vehicle have more occupants than the total seats. Traveling with an excessive number of passengers is dangerous to all occupants, even to those who may be safely restrained.

The vehicle is fitted with a driver's seat, and five or six passenger seats depending on the vehicle variant. The following are the seat adjustment options available.

First Row Seat Adjustments



1. Tilt Adjustment:

To adjust the seat surface (tilt) upward or downward, pull/push the front section of the tilt button inline Rectangle upward or downward.

2. Forwards/Rearward Adjustment:

To move the seat forward or rearward slide the travel button inline Rectangle in the intended direction of travel.

3. Seat Height:

To move the seat cushion upward or downward, pull/push the rear section of the height button inline Rectangle upward or downward as shown.

4. Seat back Angle Adjustment:

To move the backrest forward or backward, pull/push the backrest button inline Rectangle forward or backward.

5. Lumbar Support (if equipped):

The Lumbar adjustment switch consists of four sections front, rear, upper and lower. To adjust the lumbar support, press the lumbar button inline Rectangle in the desired direction.



Seat height and seat tilt adjustments are not available for the second row outboard seats for both 7-seater and 6-seater (captain's chair) variants.

Second Row Seat Adjustments



Forwards/Rearward Adjustment:

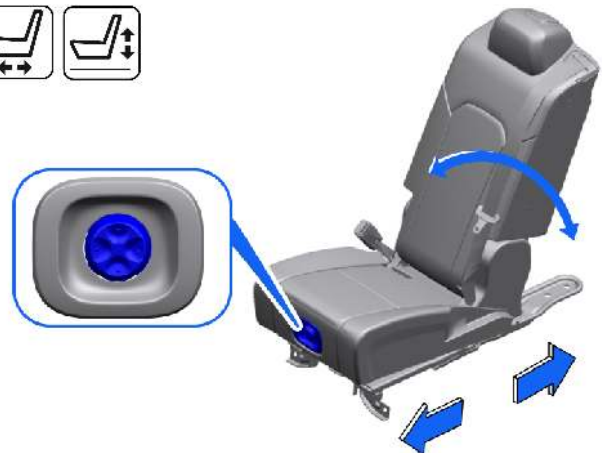
Press the upper end of the button to move the seat rearward.

Press the lower end of the button to move the seat forward.

Seat backrest Angle Adjustment:

Press the right button to move the backrest forward

Press the left button to move the backrest rearward or recline it.



Third Row Seat Adjustments

Third row seats can not be adjusted, they can only be folded down

Steering Wheel Adjustments



The steering wheel can be adjusted to by two methods. ie Electrically and Manually.

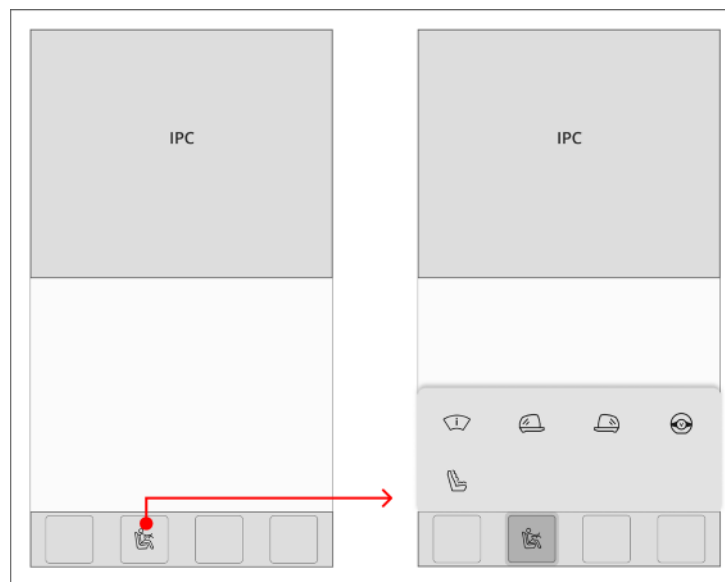


Do not make steering wheel adjustments while driving.

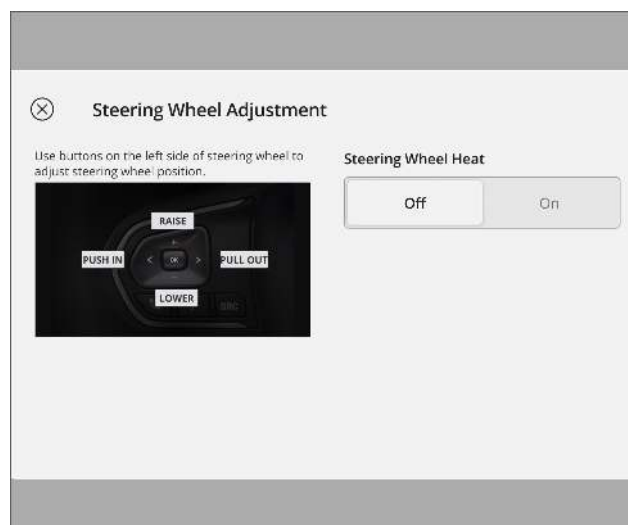
Electric Adjustable Steering Column

To adjust the steering wheel position:

- Tap Steering Adjust icon in the Car Control bar 



- To adjust steering height, press the plus [+] or minus [-] button.
- To move the steering wheel forward or back, press the left [<] or right [>] button.



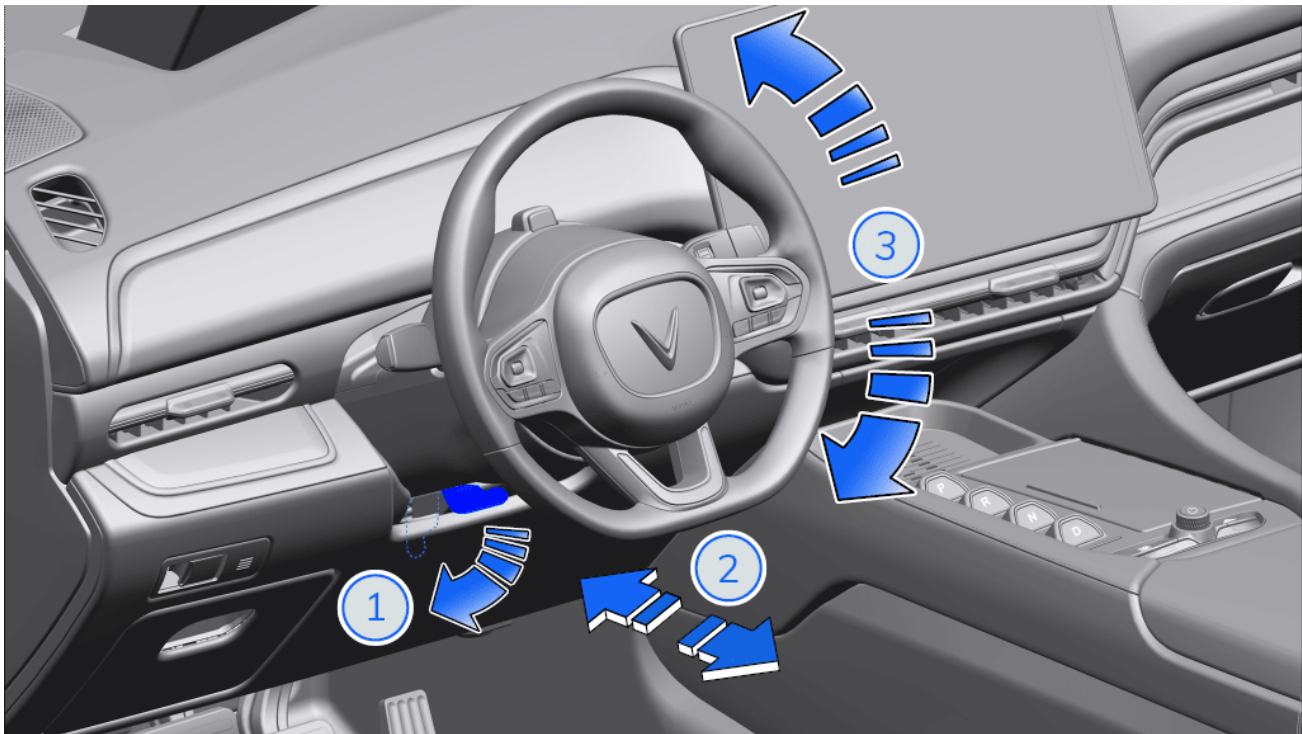
Manual Adjustable Steering Column (if equipped)

To adjust the steering wheel with manual adjustable steering column:

Unlock the steering column by releasing the lever and rotating the lever downwards. Adjust the steering wheel to your preferred position. Lock the steering wheel in the adjusted position by rotating the lever up to its original position.

Manual adjustable steering column allows for the following steering wheel adjustments.

1. Releasing Lever to make adjustments
2. fore-aft steering wheel adjustments.
3. up/down Steering wheel adjustments



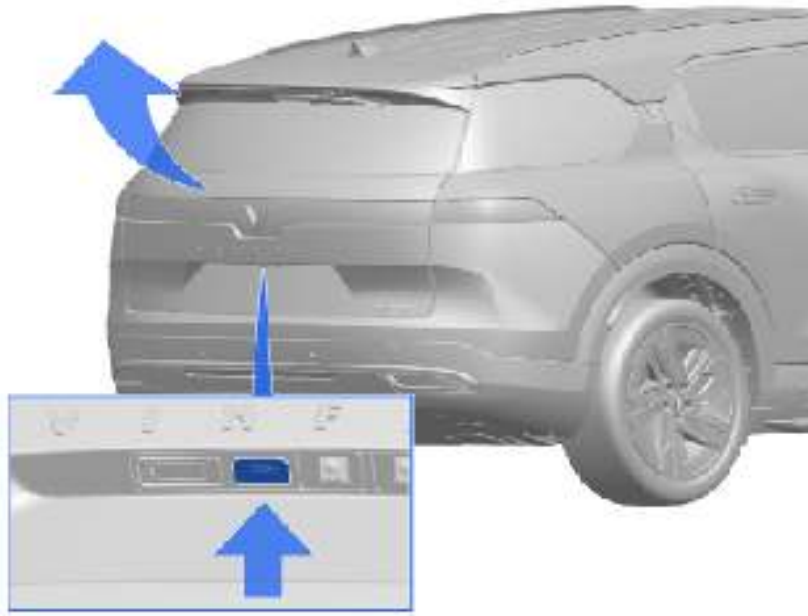
Opening the Hood

Pull the lever below the steering wheel twice to open the hood.



Opening the Tailgate

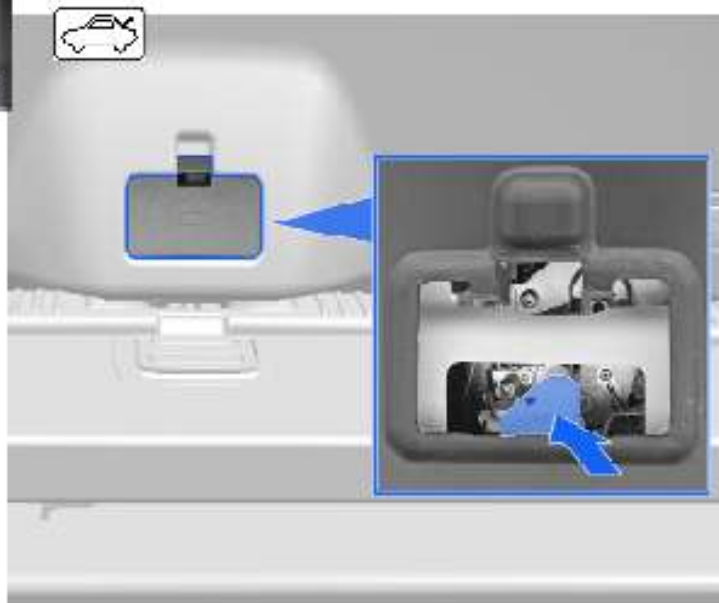
1. To open the tailgate, press the button at the base of the tailgate.



2. If no 12V power is available, the tailgate can be operated manually from inside the vehicle. To access the emergency safety release, open the inside cover on the lower side of the tailgate. Press the white tab to open the tailgate.



Tailgate view from inside the vehicle



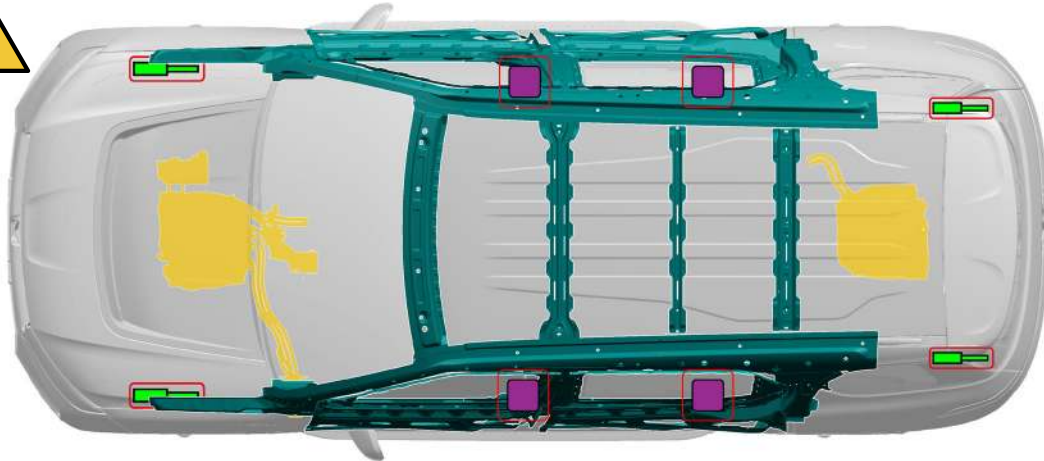
Cutting the Vehicle



Always assume that HV parts are energized. Always wear the appropriate PPE. Contact with HV current can cause serious injury or death.

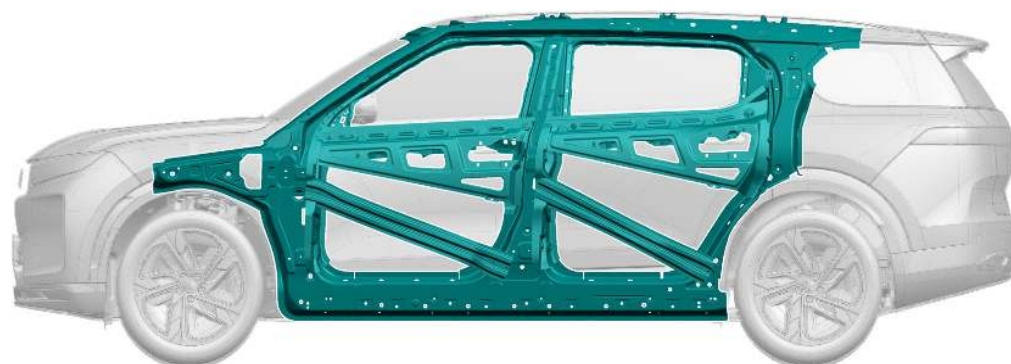
The VF9 has areas that contain HV components and other hazards that could be dangerous if a cut is made. The tailgate and the hood contains two gas struts each, that are pressurized. Dangerous cutting areas are shown in orange.

There is a risk of serious injury or death if a cut is made in these areas.



Reinforcement locations

To protect vehicle occupants, extra-high-strength steel is used. Reinforcements are shown in teal. Powerful cutting equipment must be used to cut or crush these areas.



Glass Types

A- Laminated Glass

B- Tempered Glass

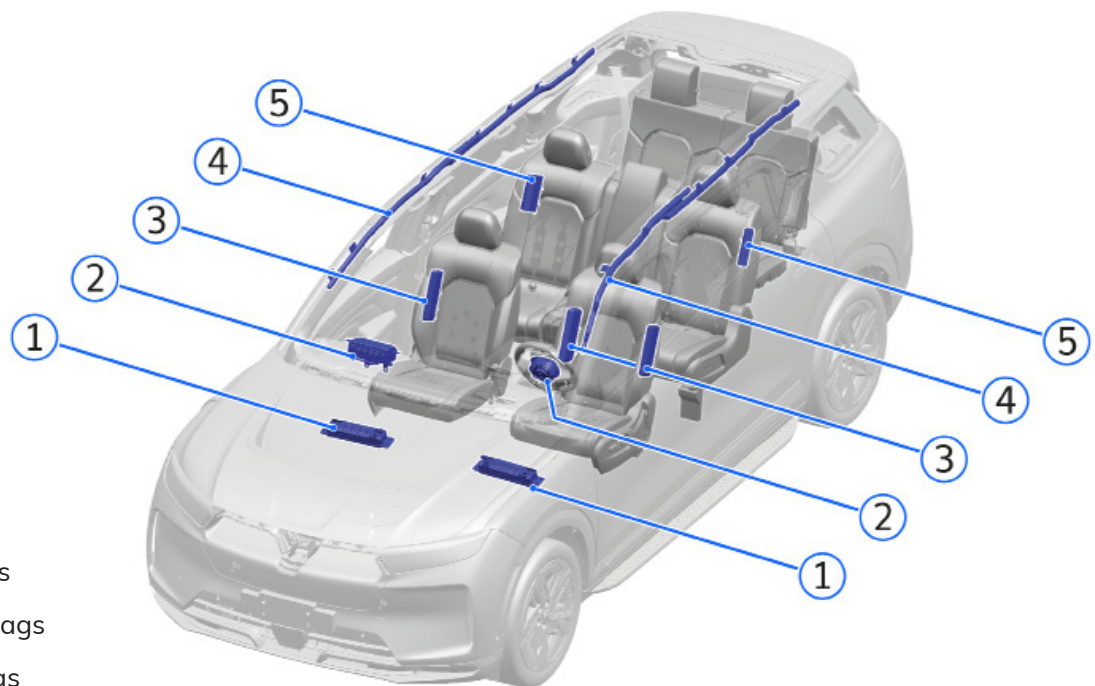


Airbags

Airbags are installed throughout the VF9, including seat-mounted airbags in the front and rear seats.

When an airbag is deployed, the vehicle will deactivate high voltage in all components and cables except the HV battery.

Air bag locations are identified below.



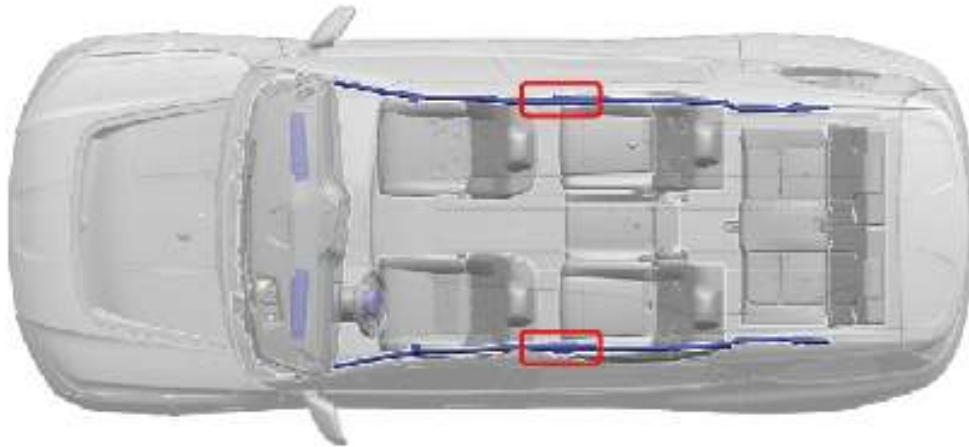
1. Knee airbags
2. Frontal airbags
3. Front side airbags
4. Curtain-airbags
5. Rear side airbag

Inflators



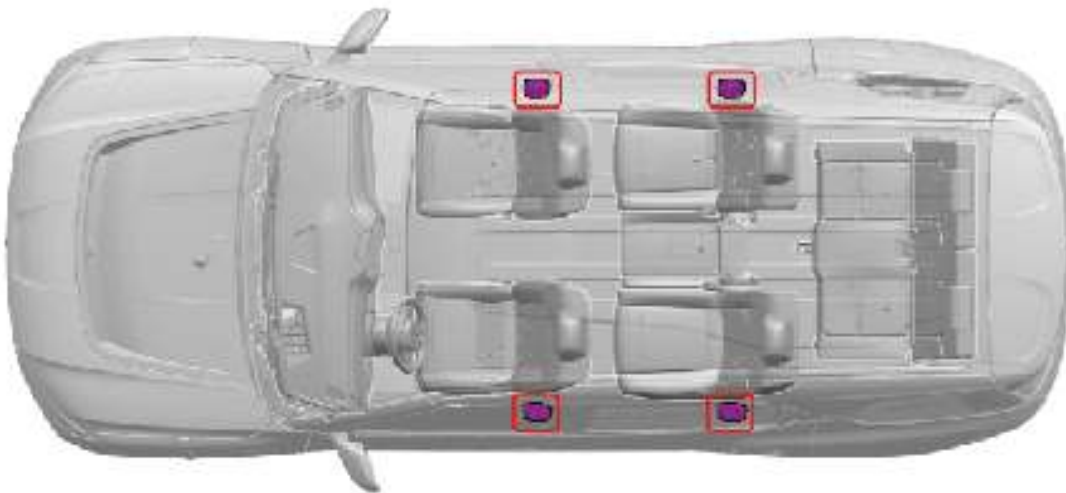
There are safety risks present when cutting the vehicle. Follow all safety precautions when working on pyrotechnic devices. Failure to comply could result in serious injury or death.

Airbag inflation cylinders are located above the rear windows. They are outlined in red.



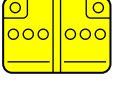
Seat Belt Pre-Tensioners

Pyrotechnic seat belt pre-tensioners are located at the bottom of the B-pillars and behind the rear seats. The seat belt pre-tensioners are outlined in red.



5. Stored energy/ liquids/ gases/ solids

Stored Energy/Liquids/Gases

	  	12V
	     	400V
	  	50% Glycol & 50% Water Mixture (15L to 18L)
	  	R1234yf (575g to 750g)

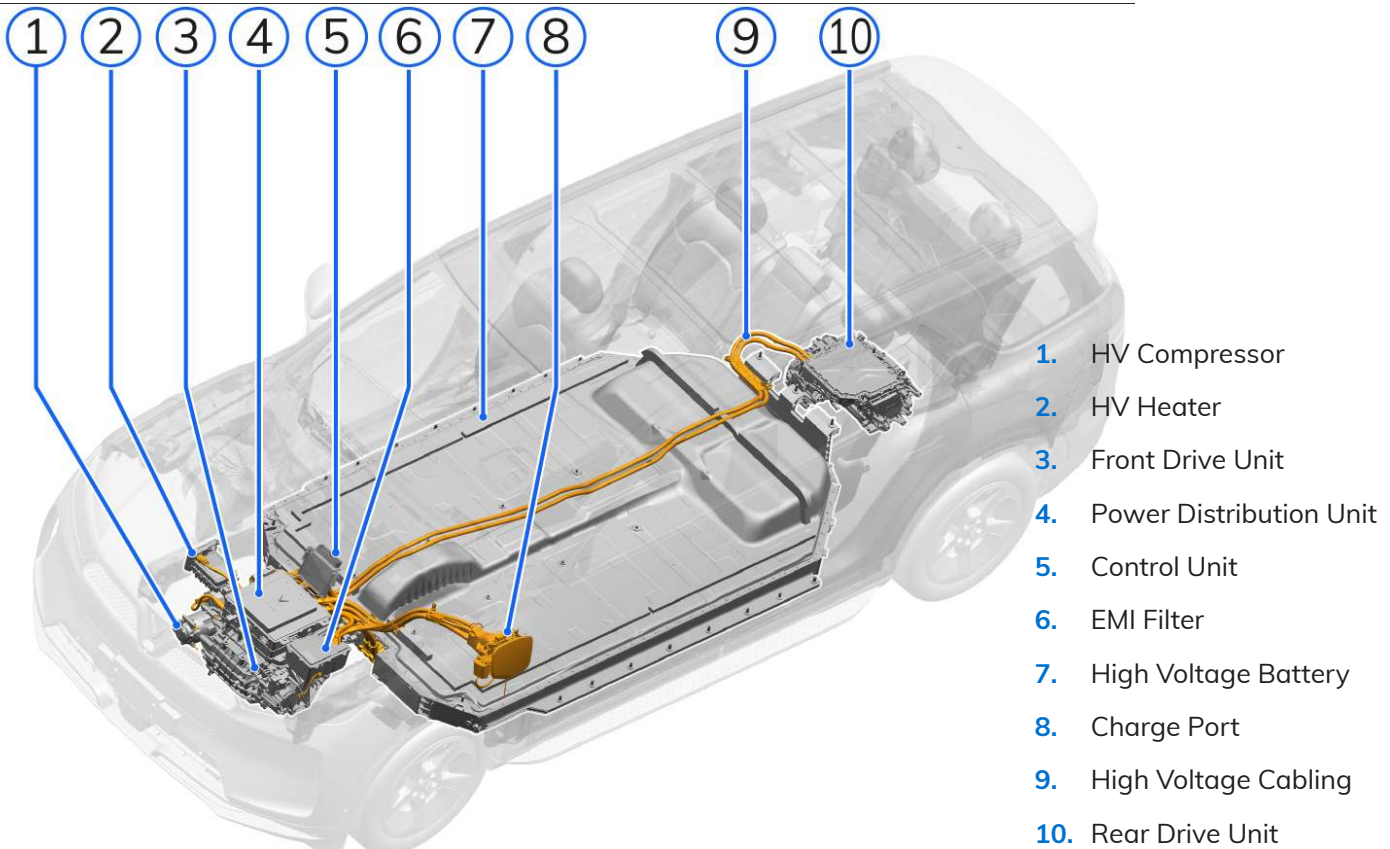
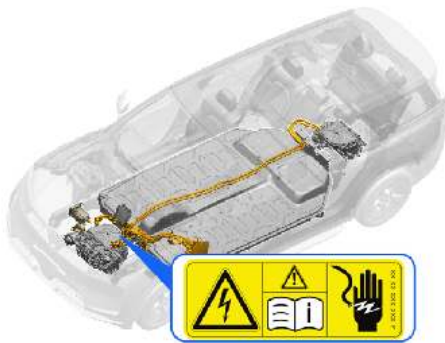
High Voltage Components



Always assume that parts are energized. Always wear the appropriate PPE. Contact with HV current can cause serious injury or death.

The VF9 contains High Voltage components. All HV components are marked with warning labels, except for HV cables. The HV cables are easily recognized by their orange warning color.

Examples of warning labels on HV components and batteries:



High Voltage Cables



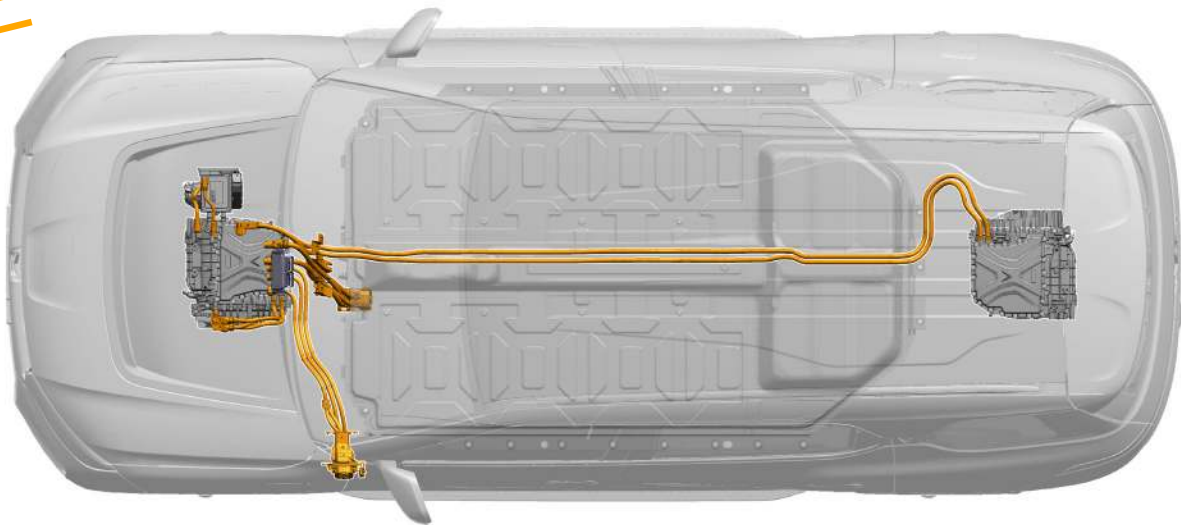
The HV system must be disabled before disconnecting any components. Never touch, cut, or disconnect the following items until the HV system is disabled:

- any orange HV cables
- any HV connectors
- any HV components

Failure to do so can result in injury or death due to electric shock.

High voltage cables are displayed in orange.

The VF9 has a first responder cut loop in the front trunk that can be cut twice to disable 12V power supply to the HV system. See [Chapter 3](#) for details.



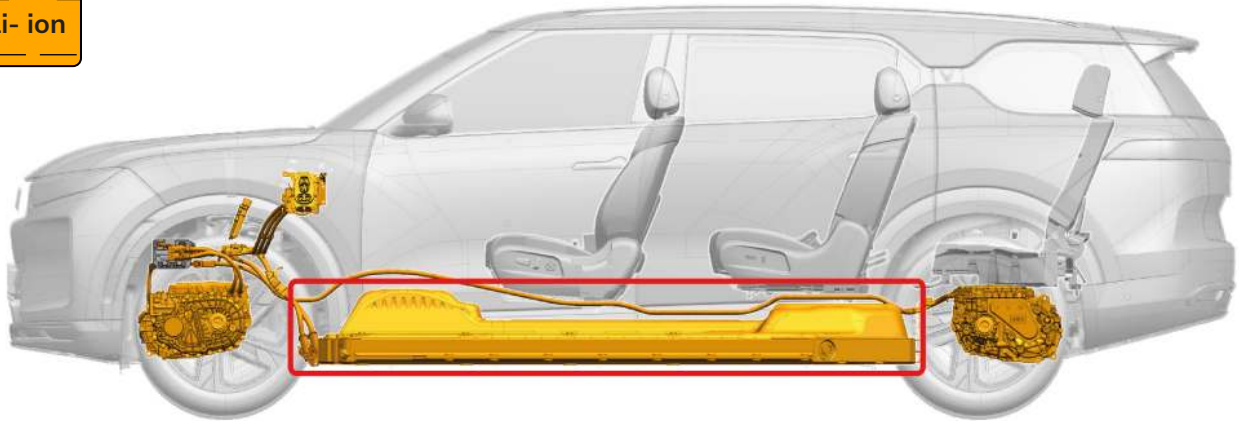
Batteries



Do not use the HV battery to lift the vehicle. The battery could become damaged. This increases the risk of injury.

High Voltage Battery

The VF9 has a 400-volt lithium-ion HV battery mounted underneath the floor pan. When using rescue equipment, make sure the floor pan is not breached.



Low Voltage (12V) Battery

The VF9 also has a low-voltage 12V battery located under the hood.

The 12V battery supplies power to the high-voltage contactors. This allows HV current to flow in and out of the HV battery. When the 12V battery is disconnected, HV current cannot flow.



6. In case of fire

Extinguishing a Fire



If the electric vehicle is on fire, it can be extinguished with large quantities of water. Use water from a hydrant or other nearby water sources whenever possible.

Lithium-ion batteries can spontaneously self-ignite. Batteries can reignite after a fire has been suppressed. This can occur several hours after an event that causes damage. Always wear the appropriate PPE and self-contained breathing apparatus during firefighting activities.



First responder:

- Provide large quantities of water to extinguish the fire
- Monitor the temperature of the battery after extinguishing
- Arrange for evacuation. Remember that the battery might reignite
- Quarantine the vehicle in case of reignition

HV battery fires:

- Flooding a battery with water can extinguish it
- If no water is available, use a dry chemical or other typical fire-extinguishing agent until water is available
- The battery can create dangerous gases when it ignites

Gaining access to battery:

- If it's determined safe to do so, raise or tilt the vehicle to gain direct access to the battery
- Do not open the battery
- Only apply water inside the battery if a natural opening exists

Reignition:

- The battery could reignite if it is placed near fire
- Check for remaining heat sources with a thermal imaging camera
- Make sure the battery is fully cooled before moving the vehicle

The HV battery system has lithium-ion cells. If this type of battery catches fire, it has a low risk of explosion because it contains an oxidizer. This type of battery will continue to burn until it is completely burned out.

7. In case of submersion

Submersion



Recovery of a submerged vehicle without appropriate PPE can result in serious injury or death.

Whether the VF9 is fully or partially submerged, the HV system does not pose an increased risk of electric shock. If the vehicle has suffered serious damage in an accident before being submerged, there could be an increased risk of electric shock during recovery. Firefighters should follow established guidelines per National Fire Protection Association (NFPA) to recover a damaged electric vehicle in a submerged state.

Remove the vehicle from the water and disable the HV system as described on [Chapter 3](#). Make sure to wear the appropriate PPE.



8. Towing/ transportation/ storage

Handling Damaged Vehicle



Monitor the battery temperature with a thermal camera (if available) after a crash to prevent thermal runaway. Thermal runaway can result in a fire or explosion as the batteries release stored energy. Battery cells can release gases in response to chemical reactions occurring inside the batteries.



Always assume that HV parts are energized. Always wear the appropriate PPE. Contact with HV current can cause serious injury or death.

If the VF9 is damaged in a crash, it should be transported to a safe storage location.

Isolate the vehicle as follows:

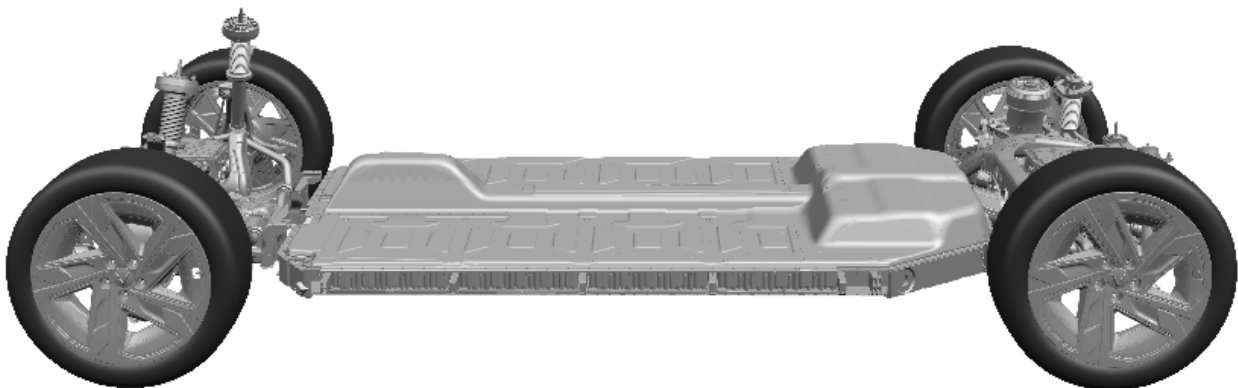
- Do not park the damaged vehicle in an enclosed building
- The damaged vehicle should be parked outdoors at a safe distance from other vehicles, buildings, and flammable objects
- Quarantine areas should be established and used whenever possible

Emergency responders must be notified of a thermal runaway. The battery pack must be cooled to stop the chemical reaction.

Do not push on the floor pan inside the VF9. When using rescue equipment, make sure the floor pan is not breached. The HV battery or the HV cables may be damaged. Both can result in serious injury or death. Wait until trained personnel arrive if floor pan is breached.

Vehicle damage

A damaged electric vehicle must not be touched without the proper PPE.



Towing



Lithium-ion batteries can spontaneously self-ignite. Batteries can reignite after a fire has been suppressed. This can occur several hours after an event that causes damage. Always wear the appropriate PPE.

The HV system must be switched off before transporting the vehicle.

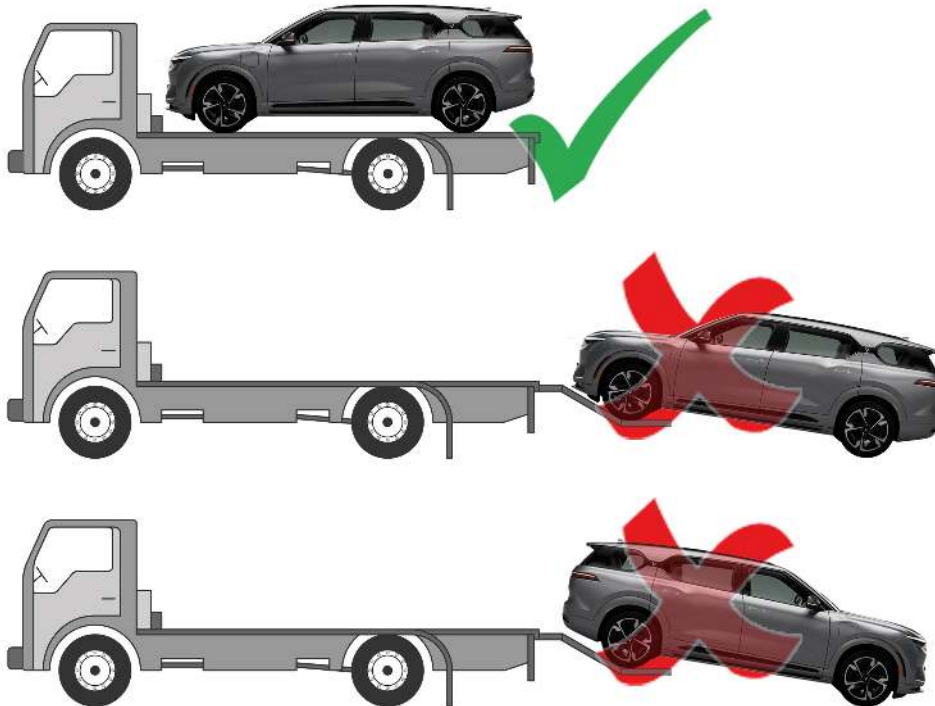
The VF9 must be fully placed on a tow truck bed for transporting. Set the gear to Neutral (N), disengage the parking brake, and turn off the vehicle.

When towed, the vehicle must not have any wheels in contact with the road surface. The wheels must be secured properly using an eight-point tie-down method.

Drive units

The VF9 has two electric drive units: front and rear. The HV battery sends direct current (DC) energy to the Electric Drive System (EDS) to rotate the wheels.

Wheel rotation can generate electricity in the EDS. This could result in overheating of HV components or cause further damage. The wheels should be chocked to prevent movement.



9. Important additional information

First Responders and Second Responders with emergencies, call VinFast Assistance Hotline.

First responder information can be found at <https://vinfastauto.eu/en>

First responders and training officers who have questions, contact: info.eu@vinfastauto.com

In case of any more information or assistance required you can contact the manufacturer. the manufacturer details are given below.

VINFAST TRADING AND PRODUCTION JOINT STOCK COMPANY

Dinh Vu - Cat Hai Economic Zone,

Cat Hai Island, Cat Hai Town,

Cat Hai District,

Hai Phong City, Vietnam

10. Explanation of pictograms used

	General Warning Sign		Flammable
	Warning, Electricity		Hazardous to the human health
	Electric Vehicle		Corrosives
	Explosives		Use Thermal Infrared Camera
	Bonnet		Use water to Extinguish the fire
	Boot		Acute Toxicity
	Air-conditioning component		Remove smart Key
	Seat height adjustment		Steering wheel, tilt control
	Seat adjustment, longitudinal		